

AP Computer Science A - Project 1.12 - 1.15

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1 Introduction

Pallotti has decided that we are going to be moving away from using PlusPortals for all of our student data, and you have been tasked with designing a new system to store all of the information that PlusPortals handled previously. The administration has asked you to create a prototype of how the individual students could be loaded into the system.

2 Students

The main feature that your prototype must include is the **Student** class. The **Student** class will be used to create individual objects for each student at Pallotti. For now, students will only be able to take 3 courses at a time. Your **Student** class should contain the following:

2.1 Attributes (Instance Variables)

Your class must define each of the following attributes:

- First name
- Last name
- Start year
- Graduation year
- Email address
- Period A course
- Period B course
- Period C course

2.2 Behaviors (Constructors and Methods)

Your class must also define each of the following methods:

- `Student(String email)`
- `Student(String firstName, String lastName, int startYear)`
- `String getEmail()`
- `int getGraduationYear()`
- `void setGraduationYear()`
- `void setCourseA(Course c)`
- `void setCourseB(Course c)`
- `void setCourseC(Course c)`
- `void printSchedule()`
- `String toString()`

3 Courses

For the prototype, students will only have access to three courses. These courses will be very simple, and they will only include data for the name of the course and the name of the teacher for that course. Your **Course** class should contain the following:

3.1 Attributes (Instance Variables)

Your class must define each of the following attributes:

- Course name
- Teacher name

3.2 Behaviors (Constructors and Methods)

Your class must also define each of the following methods:

- `Course(String courseName, String teacherName)`
- `String toString()`

4 Extension: Person

Eventually, we will need to add a way to add other people into the system, like parents and guardians or teachers. One way to do this would be to create individual classes for each type of person, but we would have a lot of duplicated code this way.

Instead, we can create a superclass **Person** with general personal information and methods, and then have our students, teachers, and parents all inherit from that superclass. This means that all of the subclasses would not need to redeclare or redefine the base methods and attributes from the superclass.

To achieve this inheritance, we write the superclass just like normal, and then have our subclasses inherit from the superclass using the **extends** keyword.

For example, if we create a **Person** class and have our **Student** class, we can make **Student** a subclass **Person** by changing the first line of the `Student.java` file to:

```
public class Student extends Parent
```

Use inheritance to create a generic **Person** class that will handle the names and email of each person. Then, modify the **Student** class so that it inherits from the **Person** class, rather than defining its own names and email. Finally, use the **Person** class to add a guardian attribute to the **Student** class, and replace the teacher name attribute of **Course** with an instance of the **Person** class.

4.1 Attributes (Instance Variables)

Your class must define each of the following attributes. These must be defined as **protected** visibility, otherwise these will not be accessible in subclasses. Note the overlap with the **Student** class:

- First name
- Last name
- Email address

4.2 Behaviors (Constructors and Methods)

Your class must also define each of the following methods:

- `Person(String firstName, String lastName)`
- `String getFullName()`
- `String getEmail()`
- `String toString()`